

September 23, 2026

COMP440 Game Design

Project 2 Checkpoint: Game Design Document **Heirloom**

Daniel A.

a. **Core Loop** in one sentence.

The player must escape their cursed family home before dawn, or they will be trapped in the house, and the next generation will be cursed with a stronger version of the curse.

Samaii H.

b. **The Three Systems**

Time System — Owner: **Lionel B.**

Owns the game timer, time remaining, and loop/reset state. It controls the countdown to dawn and determines when the player runs out of time and the loop restarts.

Inventory System — Owner: **Daniel A.**

Owns the player's current heirloom, collected items, and item effects. It controls picking up, dropping, switching, and using heirlooms.

Environment/Puzzle System — Owner: **Samaii H.**

Owns puzzle states, locked/unlocked areas, hidden objects, and environmental interactions. It controls how the house changes based on the player's actions, items, and progress.

c. **The Seams**

One seam is **ActiveHeirloom**. What crosses is the name or type of heirloom the player currently has equipped. Daniel A. writes it through the Inventory System, and Samaii H. reads it through the Environment/Puzzle System. This allows puzzles and hidden objects to react to specific heirlooms.

Another seam is **TimeRemaining**. What crosses is the amount of time left before dawn and whether the loop has ended. Lionel writes it through the Time System, and Samaii H. reads it through the Environment/Puzzle System. This allows the environment to change as time runs out and reset when the timer reaches zero.

A third seam is **TimeEffect**. What crosses is whether the player is using an heirloom that affects time, such as the Grandfather's Watch. Daniel A. writes it through the

Inventory System, and Lionel reads it through the Time System. This allows the timer to slow down or change while certain heirlooms are active.

Lionel B.

d. Your Reading of Inheritance

Our understanding of inheritance is that people inherit both the benefits and consequences of their family. The player inherits a home, but they also inherit the curse, memories, abilities, and the unfinished mission of the generations that came before them. If the player is unable to finish the mission before dawn, the curse will go to the next generation, leaving someone else to inherit the house and complete the unfinished mission.

Danieal A.

e. Art Direction:

1. **The world, in one sentence.** Not the plot. The place, its condition, and how it got that way.
 - a. An old-fashioned, old-money family mansion that was cursed decades ago sits alone in a cul-de-sac. The inside of the home is fancy, with its brown checkered marble floors and statues, but it's dark, gloomy, and cold; yet it's clean, since cleaning is one of the only things the trapped inhabitants could do to pass the time. The second floor, though, is a different story; it's unkempt, ridden with spider webs, and cold, as though the residents were too scared to go up there...
2. **Palette.** Five colors with hex values. One is the accent, and it appears in almost nothing.
 - a. #261809

- b. #e5e8e9
 - c. #000000
 - d. #b67433
 - e. #301703
3. **Camera and framing.** What angle is everything viewed from, and at what apparent scale?
- a. The game world is viewed from a third person point of view for cutscenes but the game is played in a third person POV at normal scale with a slight zoom out and slight distortion at the edges of the screen during supernatural encounters.
 - b. For the cover, there should be a young girl standing right inside the front door with it wide open (opening inwards) and the clean, checkered, marble floors are shining with statues sitting on top of them and a fountain in the middle of the room behind her while the stairs to the second floor are ascendingly beat up. The second floor is much darker, but we can see the railing going from left to right (from outside still) and the large spiderwebs covering the doors up there. The girl stands at the front door adorned in a white dress that is only dirty at the bottom, and she smiles with her head tilted to the left.
4. **Light.** Direction, hardness, temperature, and whether there is any fill.
- a. The 1st floor of the mansion is dimly lit with chandeliers, while the second floor is nearly pitch black.
 - b. Outside, the skies are clear and blue except for an entire strip of sky above the house. The long strip of sky above the mansion is gloomy and cloudy; it looks like it's about to rain and goes on for miles, but the width is only the size of the James family compound.
5. **Edge and detail.** How does an object meet the background, and where is detail allowed to live?
- a. We see the back of someone's shoulder and head as they stand in the middle of the cul-de-sac, staring at the little girl in the doorway of her home, smiling. Detail is focused on the family home and the strip of gloomy sky that lies above it. The person staring also stands under the strip associated with the mansion.
6. **Mood, in three words.** Three, not five. Specific enough that two of your classmates could not have written the same three.

- 1. Mysterious, Eerie, Charged

Lionel B.

- f. **Scope** What is in, and three cuts. At least one must be a fourth system.

The game will include heirlooms with abilities and also downsides to go alongside them. Puzzles to unlock new areas of the house and a countdown until dawn

Cut 1 - Farming System

- A complete farming system would be our fourth system and our first major cut. The farm can still exist behind the house and story and exploration, but planting crops, harvesting food, hunger, and resource management will be removed if the project becomes too large.

Cut 2 - Dracula Boss Fight

- The full Dracula boss fight can be removed if combat takes too much development time. Dracula could instead become a simpler enemy, a quick encounter or an escape sequence.

Cut 3 - Additional Enemies

- The number of enemies could be reduced. We will prioritize enemies that interact with our main systems, such as the Maid puzzle pieces or Cursed Dolls interfering with the player's item. Other enemies can be added if there is enough development time.

Samaii H.

- g. **Which Seam Breaks First**

The seam most likely to break first is **ActiveHeirloom** because multiple systems depend on knowing which heirloom the player currently has equipped. The Inventory System writes the active heirloom value when the player switches items, while the Environment/Puzzle System reads it to decide which abilities, hidden objects, or puzzle interactions should activate. If the value does not update correctly, the game may use the wrong heirloom effect, fail to reveal objects, or block a puzzle even when the player has the correct item equipped.